

Communication Diagram

Communication Diagram

- Communication diagrams, a kind of interaction diagram, emphasize the data links between the various participants in the interaction.
- UML Communication Diagrams, previously known as collaboration diagrams are a type of behavioral diagram that shows the interactions that take place between objects in a piece of software or system.
- This type of diagram emphasizes the messages exchanged between objects. Communication diagrams are best used when **one use case has multiple scenarios** that need depicting.

Communication & Sequence Diagram

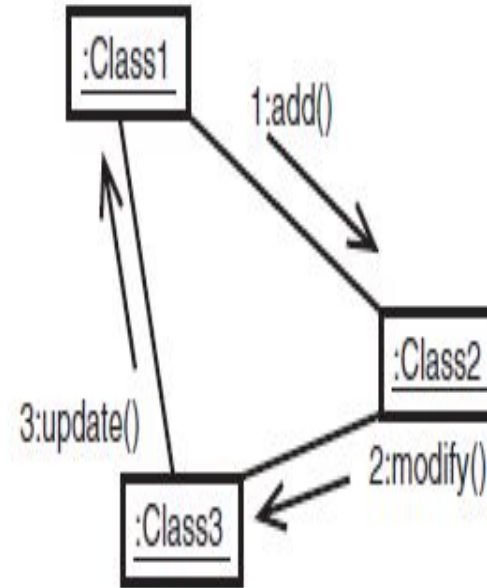
- Sequence and Communication diagram shows how objects interact.
- It shows objects along with the messages that travel from one object to the another.
- Sequence diagram shows the time ordering of interaction
- Communication diagram emphasizes the **context** and overall organization.

Symbols

- Rectangles represent objects that make up the application.
- Lines between class instances represent the relationships between different parts of the application.
- Arrows represent the messages that are sent between objects.
- Numbering lets you know in what order the messages are sent and how many messages are required to finish a process.

FIGURE 10.1

The symbol set for the communication diagram.



Object and Communication Diagram

- Object Diagram shows objects and their relationships with one another.
- Communication Diagram is extension of Object Diagram.
- Object Diagram is like a snapshot and Communication is like a movie.
- Object Diagram shows instances and their linkage.
- Communication shows the message among instances over the time.

Object & Communication Diagram

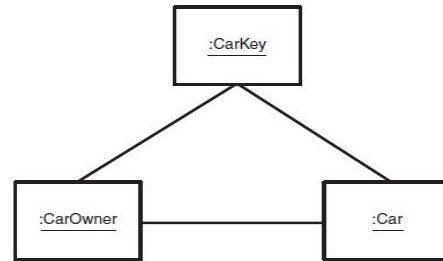
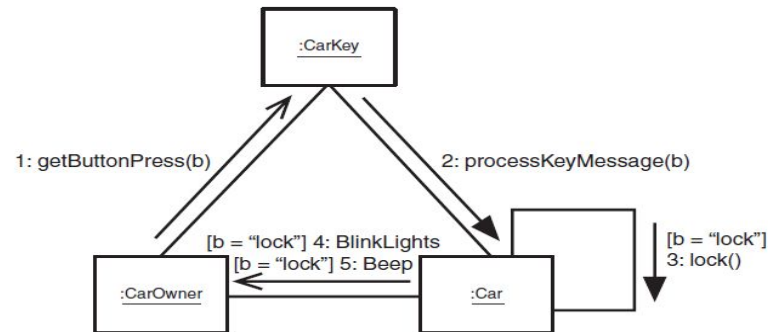


FIGURE 10.3
An object diagram
that models
instances of the
classes in
Figure 10.2.

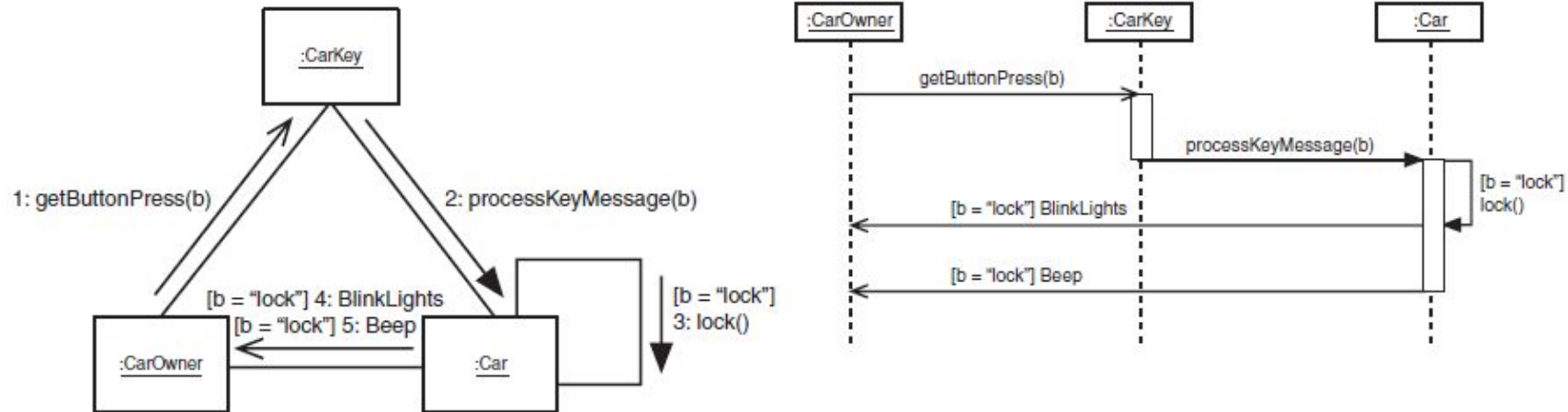
FIGURE 10.4
A communication
diagram that
models the
messages that
pass between the
objects in
Figure 10.3



Communication & Sequence are semantically Equivalent:

FIGURE 10.5

The communication diagram and the equivalent sequence diagram for the car and car key example.



Communication Diagram Example:

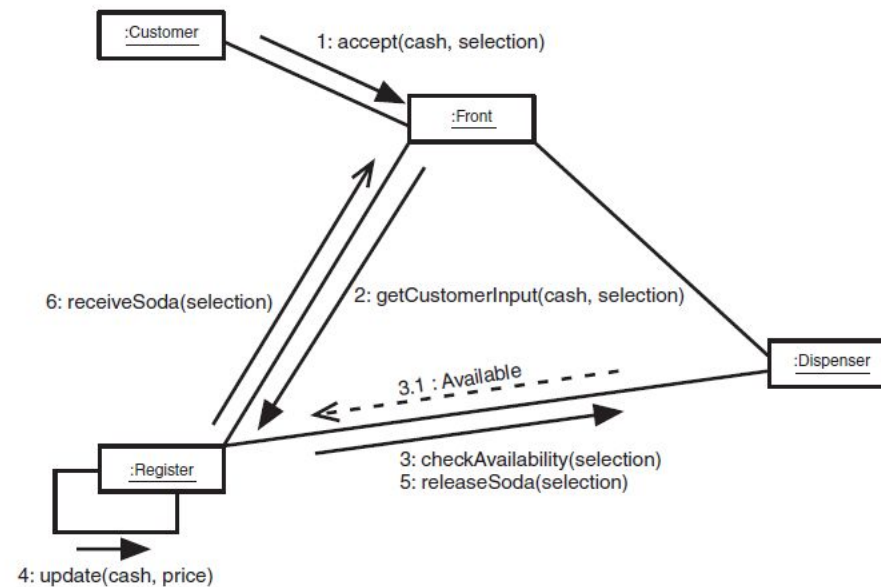
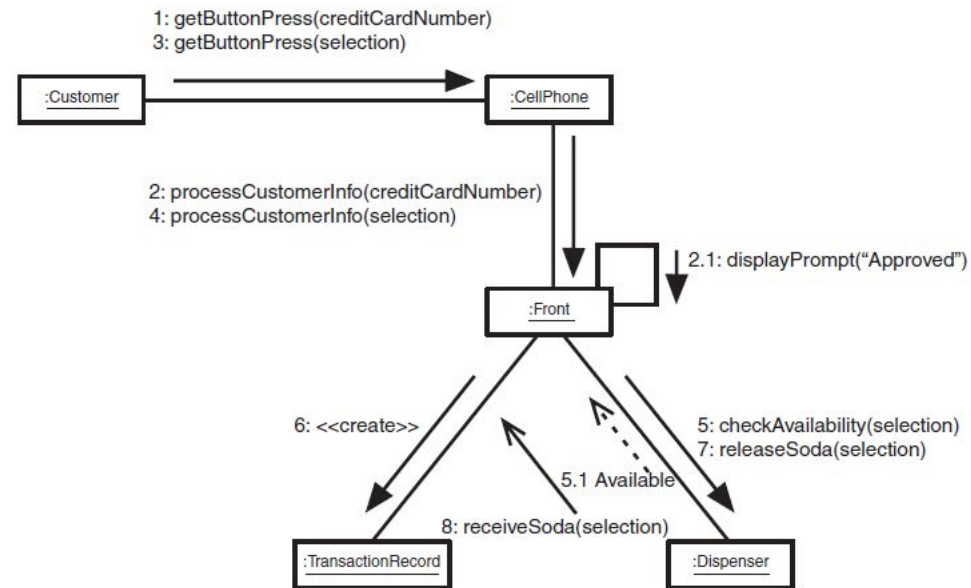


FIGURE 10.9

The communication diagram for the best-case scenario of the “Buy soda” use case.

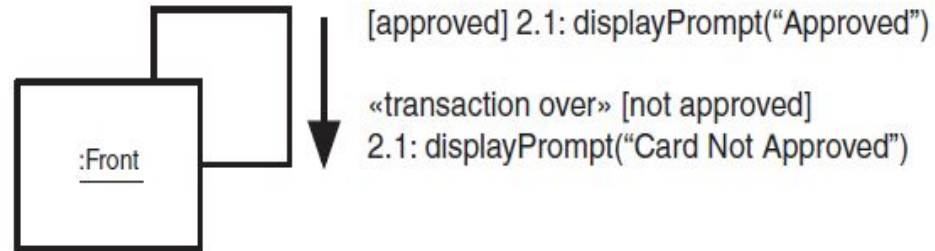
Creating an Object

FIGURE 10.11
Modeling object
creation in the
best-case
scenario in the
cell-phone-enabled
soda machine.



Numbering the messages

FIGURE 10.12
Numbering the
messages for
mutually exclusive
guard conditions.



Multiple Receiving Objects

- Sometimes an object send message to multiple objects

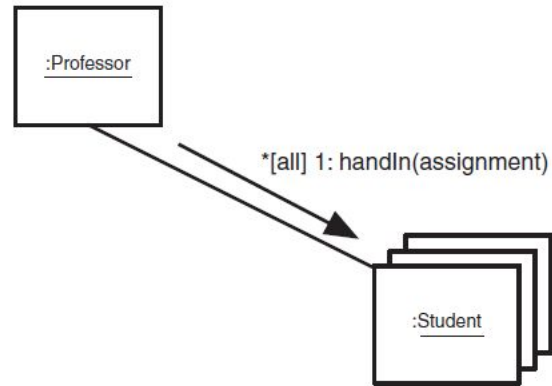


FIGURE 10.13

An object sending a message to multiple objects in a class.

Order of message for multiple Objects

- Order of message is important

For Example: A bank serves each customer in the order they appear in line.

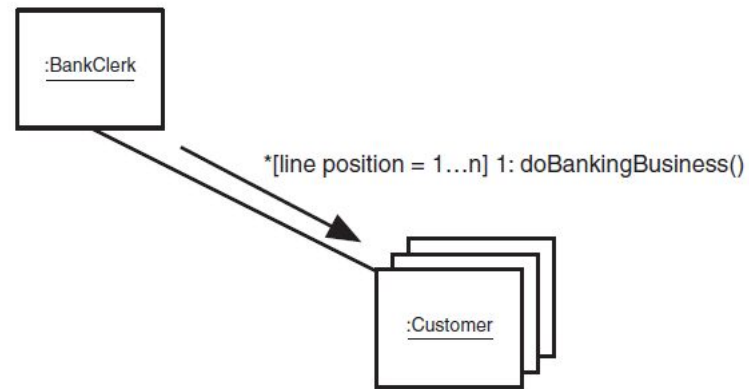


FIGURE 10.14
An object sending a message in a specified order to multiple objects in a class.

- A message can be request for an object to perform calculation and return a value

FIGURE 10.15
A communication
diagram that
includes the syntax
for a returned
result.

